

Expectation

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A random variable and its distribution

A random variable X assigns a number $X(\omega)$ to an outcome ω in a sample space Ω .

$$p_X(x) = P(X = x) = \sum_{\omega: X(\omega)=x} p(\omega)$$

is its probability mass function. For two fair, independent coin flips, let X count heads. With H for heads and T for tails,

$$\Omega = \{HH, HT, TH, TT\}, \quad p(\omega) = 1/4.$$

x	0	1	2
$p_X(x)$	1/4	1/2	1/4

We can use this distribution to calculate averages and variability.

Expected value

Let S_X be the set of possible values of X . The **expected value**, or **mean**, is the probability-weighted average

$$\mathbb{E}[X] = \sum_{x \in S_X} x p_X(x).$$

For infinitely many values, we assume $\sum_x |x| p_X(x) < \infty$ when using a finite expectation. For the number of heads in two fair flips,

$$\mathbb{E}[X] = 0 \left(\frac{1}{4} \right) + 1 \left(\frac{1}{2} \right) + 2 \left(\frac{1}{4} \right) = 1.$$

The weights are probabilities, not equal weights for the distinct values.

An expectation need not be a possible value

Let D be the result of a fair die roll. Then

$$\mathbb{E}[D] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5.$$

The die never shows 3.5. An expectation is not a prediction of the exact next outcome.

It describes an average. Over many independent repetitions of the same experiment, sample averages tend to be close to the expectation (under the assumptions of the law of large numbers). The most probable value and the expected value need not coincide either.

A game with an entry fee

Problem

A game costs two tokens to play. Roll a fair die once. You receive six tokens if the result is six, and nothing otherwise.

What is your expected net gain per play? What entry fee would make the expected net gain zero?

The game: solution

Model. $\Omega = \{1, \dots, 6\}$, with $p(i) = 1/6$. Let G be net gain, after subtracting the two-token entry fee:

$$G(i) = \begin{cases} 4 & i = 6, \\ -2 & i \in \{1, \dots, 5\}. \end{cases}$$

The events $\{G = 4\} = \{6\}$ and $\{G = -2\} = \{1, \dots, 5\}$ give

$$P(G = 4) = 1/6, \quad P(G = -2) = 5/6.$$

$$\mathbb{E}[G] = 4 \left(\frac{1}{6} \right) - 2 \left(\frac{5}{6} \right) = -1.$$

The expected loss is one token per play, although a single play may win. The expected payout is $6(1/6) = 1$, so an entry fee of one token makes the expected net gain zero.

Functions of a random variable

If we record $g(X)$ instead of X , its expectation is

$$\mathbb{E}[g(X)] = \sum_x g(x)p_X(x).$$

In particular,

$$\mathbb{E}[X^2] = \sum_x x^2 p_X(x).$$

For the number of heads in two fair flips,

$$\mathbb{E}[X^2] = 0^2 \left(\frac{1}{4}\right) + 1^2 \left(\frac{1}{2}\right) + 2^2 \left(\frac{1}{4}\right) = \frac{3}{2}.$$

This differs from $(\mathbb{E}[X])^2 = 1$. In general, applying a function and taking an expectation do not commute.

Counting red balls

Problem

Select two balls without replacement from a bag containing three red balls and two blue balls, uniformly at each draw. What is the expected number of red balls selected?

Counting red balls: solution

Label the balls by $L = \{r_1, r_2, r_3, b_1, b_2\}$. The outcomes are ordered pairs of distinct balls:

$$\Omega = \{(u, v) \in L^2 : u \neq v\}, \quad p((u, v)) = 1/20.$$

Let $X(u, v)$ be the number of red balls in the pair. The possible values of X are 0, 1, 2.

$$p_X(0) = \frac{2 \cdot 1}{20} = \frac{1}{10} \quad (\text{two blue}),$$

$$p_X(1) = \frac{3 \cdot 2 + 2 \cdot 3}{20} = \frac{6}{10} \quad (\text{red-blue or blue-red}),$$

$$p_X(2) = \frac{3 \cdot 2}{20} = \frac{3}{10} \quad (\text{two red}).$$

By the definition of expectation,

$$\mathbb{E}[X] = \sum_{x=0}^2 x p_X(x) = 0 \cdot \frac{1}{10} + 1 \cdot \frac{6}{10} + 2 \cdot \frac{3}{10} = \boxed{\frac{6}{5}}.$$

Adding random variables

Roll two fair dice independently. Use

$$\Omega = \{(i, j) : i, j \in \{1, \dots, 6\}\}, \quad p((i, j)) = 1/36.$$

Let $X(i, j) = i$ and $Y(i, j) = j$ record the first and second results.

Define $Z = X + Y$ by adding their values at the **same outcome**:

$$Z(i, j) = X(i, j) + Y(i, j) = i + j.$$

Thus Z is another random variable: the sum of the two results. For outcome $(2, 5)$, $X = 2$, $Y = 5$, and $Z = 7$.

More generally, for constants a, b and variables on the same sample space, define

$$(aX + bY)(\omega) = aX(\omega) + bY(\omega).$$

How is the expectation of this new variable related to $\mathbb{E}[X]$ and $\mathbb{E}[Y]$?

Linearity of expectation

For random variables X, Y with finite expectations and constants a, b ,

$$\boxed{\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y].}$$

On a finite sample space with outcome probabilities $p(\omega)$,

$$\begin{aligned}\mathbb{E}[aX + bY] &= \sum_{\omega \in \Omega} (aX(\omega) + bY(\omega))p(\omega) \\ &= a \sum_{\omega \in \Omega} X(\omega)p(\omega) + b \sum_{\omega \in \Omega} Y(\omega)p(\omega).\end{aligned}$$

Independence is not required. The same rule extends to any finite sum of random variables.

Indicator variables

For an event A , its **indicator variable** is

$$I_A = \begin{cases} 1 & \text{if } A \text{ occurs,} \\ 0 & \text{otherwise.} \end{cases}$$

Its expected value is particularly simple:

$$\mathbb{E}[I_A] = 1 \cdot P(A) + 0 \cdot P(A^c) = P(A).$$

If a quantity counts how many events occur, write it as a sum of indicators. Then add their expectations.

The mean of a binomial random variable

Let $X \sim \text{Binomial}(n, p)$. For trial i , let $I_i = 1$ if it succeeds and $I_i = 0$ otherwise.

$$X = I_1 + \cdots + I_n, \quad \mathbb{E}[I_i] = p.$$

By linearity,

$$\mathbb{E}[X] = \sum_{i=1}^n \mathbb{E}[I_i] = np.$$

For five users with click probability 0.2, the expected number of clicks is $5(0.2) = 1$.

The expectation calculation only needs the individual success probabilities; the binomial distribution additionally needs independence.

Expectation as a sum of tail probabilities

Let T take positive integer values. By definition,

$$\mathbb{E}[T] = \sum_{t=1}^{\infty} tP(T = t).$$

Now expand the probabilities of exceeding $0, 1, 2, \dots$:

$$\begin{array}{rcll} P(T > 0) & = & P(T = 1) & + P(T = 2) & + P(T = 3) & + \dots \\ P(T > 1) & = & & P(T = 2) & + P(T = 3) & + \dots \\ P(T > 2) & = & & & P(T = 3) & + \dots \\ & \vdots & & & & \end{array}$$

Adding these rows counts $P(T = t)$ exactly t times, once for each $m = 0, \dots, t - 1$. Therefore,

$$\boxed{\sum_{m=0}^{\infty} P(T > m) = \sum_{t=1}^{\infty} tP(T = t) = \mathbb{E}[T].}$$

All terms are nonnegative, so regrouping the sums is valid.

The mean waiting time

Let $T \sim \text{Geometric}(p)$, with $0 < p < 1$. T counts attempts up to and including the first success.

The event $T > m$ means that the first m attempts all fail.

Independence gives

$$P(T > m) = (1 - p)^m, \quad m = 0, 1, 2, \dots$$

For $m = 0$, $P(T > 0) = 1$ because at least one attempt is needed.

Using the formula just derived and the geometric series,

$$\begin{aligned}\mathbb{E}[T] &= \sum_{m=0}^{\infty} P(T > m) = \sum_{m=0}^{\infty} (1 - p)^m \\ &= \frac{1}{1 - (1 - p)} = \boxed{\frac{1}{p}}.\end{aligned}$$

At $p = 0.2$, the expected number of attempts is five. This does not guarantee success by attempt five.

The same mean, different variability

Compare two rewards:

- ▶ X : always receive five tokens.
- ▶ Y : receive zero or ten tokens, each with probability $1/2$.

$$\mathbb{E}[X] = 5, \quad \mathbb{E}[Y] = 0 \cdot \frac{1}{2} + 10 \cdot \frac{1}{2} = 5.$$

The means are equal, but X never varies. Every value of Y is five tokens away from its mean.

How can we measure this difference in spread?

Variance

Let $\mu = \mathbb{E}[X]$. The **variance** of X is

$$\text{Var}(X) = \mathbb{E}[(X - \mu)^2] = \sum_x (x - \mu)^2 p_X(x).$$

We assume the second moment $\mathbb{E}[X^2]$ is finite. Variance is nonnegative and measures spread around the mean. For the previous example,

$$\text{Var}(X) = 0, \quad \text{Var}(Y) = 25 \left(\frac{1}{2}\right) + 25 \left(\frac{1}{2}\right) = 25.$$

The **standard deviation** is $\sqrt{\text{Var}(X)}$. Unlike variance, it has the same units as X .

Computing variance

Expand the square and use linearity of expectation:

$$\begin{aligned}\mathrm{Var}(X) &= \mathbb{E}[X^2 - 2\mu X + \mu^2] \\ &= \mathbb{E}[X^2] - 2\mu\mathbb{E}[X] + \mu^2 \\ &= \boxed{\mathbb{E}[X^2] - (\mathbb{E}[X])^2}.\end{aligned}$$

For two fair flips, we already found $\mathbb{E}[X] = 1$ and $\mathbb{E}[X^2] = 3/2$. Thus

$$\mathrm{Var}(X) = \frac{3}{2} - 1^2 = \frac{1}{2}.$$

This identity is derived from the definition, not a separate assumption.

Bernoulli variance

Let $I \sim \text{Bernoulli}(p)$. Because I is either 0 or 1, we have $I^2 = I$.

$$\mathbb{E}[I^2] = \mathbb{E}[I] = p.$$

Therefore

$$\text{Var}(I) = p - p^2 = p(1 - p).$$

At $p = 0$ or $p = 1$ there is no variability. The variance is largest at $p = 1/2$, since $p(1 - p) = 1/4 - (p - 1/2)^2$.

Shifting and scaling

Let $\mu = \mathbb{E}[X]$, and let a, b be constants. Linearity gives $\mathbb{E}[aX + b] = a\mu + b$. By the definition of variance,

$$\begin{aligned}\text{Var}(aX + b) &= \mathbb{E}[(aX + b - (a\mu + b))^2] \\ &= \mathbb{E}[a^2(X - \mu)^2] \\ &= \boxed{a^2 \text{Var}(X)}.\end{aligned}$$

Adding a constant shifts the values but does not change their spread. Multiplying all values by two multiplies variance by four, and standard deviation by two.

Variance of a sum

Discrete random variables X, Y are **independent** if

$$P(X = x, Y = y) = P(X = x)P(Y = y) \quad \text{for every } x, y.$$

Here the comma means that both equalities hold. For independent variables with finite second moments, the following result holds (stated without proof):

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y).$$

Expectations add without independence:

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y].$$

Variances add for independent variables, but not in general.

For example, $X + X = 2X$ has variance $4 \text{Var}(X)$, not $2 \text{Var}(X)$, unless $\text{Var}(X) = 0$.

Binomial variance

For $X \sim \text{Binomial}(n, p)$, write $X = I_1 + \cdots + I_n$ using the independent trial indicators. Their variances add:

$$\text{Var}(X) = \sum_{i=1}^n \text{Var}(I_i) = np(1-p).$$

With $n = 5$ and $p = 0.2$,

$$\mathbb{E}[X] = 1, \quad \text{Var}(X) = 0.8, \quad \text{standard deviation} = \sqrt{0.8} \approx 0.894.$$

The mean describes the center; variance describes the spread. Neither alone specifies the entire distribution.

Averages vary less

Let $X_1, \dots, X_n$ be independent observations with common mean μ and standard deviation σ , hence variance σ^2 . Their sample mean is

$$\bar{X} = \frac{X_1 + \dots + X_n}{n}.$$

Linearity gives $\mathbb{E}[\bar{X}] = \mu$. Since $\text{Var}(aX) = a^2 \text{Var}(X)$,

$$\text{Var}(\bar{X}) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}.$$

The standard deviation of the average is $\sigma/\sqrt{n}$. This explains one benefit of collecting more independent observations.

Estimating a click probability

Problem

Under independent trials with a common click probability, estimate that probability by the fraction of users who click.

How does using 400 users instead of 100 affect the variability of the estimate?

The sample proportion: solution

For n users, use $\Omega = \{0, 1\}^n$ and $I_i(\omega) = \omega_i$. Let the common click probability be p , with $0 < p < 1$. The observed proportion is $\hat{p} = (I_1 + \cdots + I_n)/n$. From the Bernoulli mean and variance,

$$\mathbb{E}[\hat{p}] = p, \quad \text{Var}(\hat{p}) = \frac{p(1-p)}{n}.$$

For example, if $p = 0.2$, the standard deviations are

$$\sqrt{\frac{0.2(0.8)}{100}} = 0.04, \quad \sqrt{\frac{0.2(0.8)}{400}} = 0.02.$$

Four times as many independent observations halve the standard deviation. They do not eliminate sampling error.

Expectation and variability

- ▶ Expectation is a probability-weighted average, not a promised outcome.
- ▶ Expectations of sums always add when the expectations are finite.
- ▶ Variance measures squared deviations from the mean.
- ▶ Independent random variables have additive variances.

	$\mathbb{E}[X]$	$\text{Var}(X)$
$X \sim \text{Bernoulli}(p)$	p	$p(1 - p)$
$X \sim \text{Binomial}(n, p)$	np	$np(1 - p)$

A complete distribution contains more information than its mean and variance.